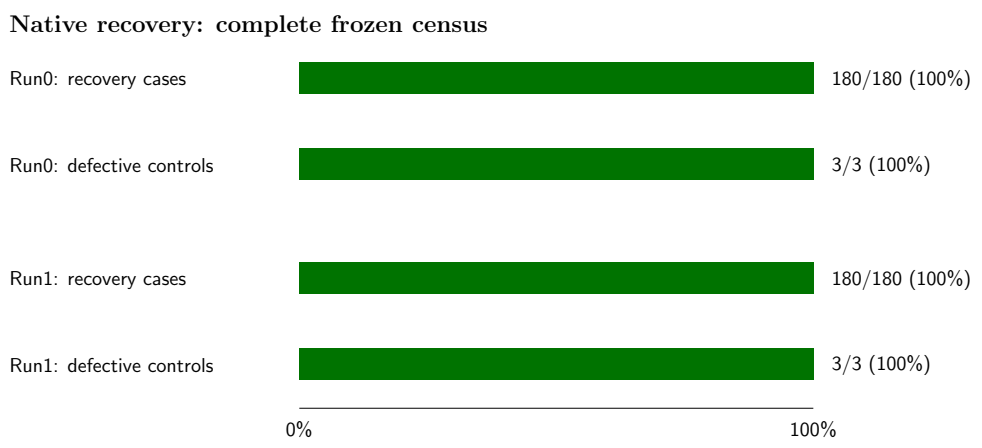
Native rational Hermite recovery

November9,2026. Study prime-hermite-native-20261109, protocolv2. Producer claim PCC-20261109-4d31b7; unsigned [pcc:unverified].

Observed result. Both physical invocations completed180 positive cases and detected3 deliberate defects. The separately frozen native analyzer accepted all366 records, checked complete unique-key coverage and exact integrality/congruence, and required byte-identical replay.

Execution	Positive correct	Defects detected	Exit
native-census/0	180/180	3/3	0
replay/0	180/180	3/3	0

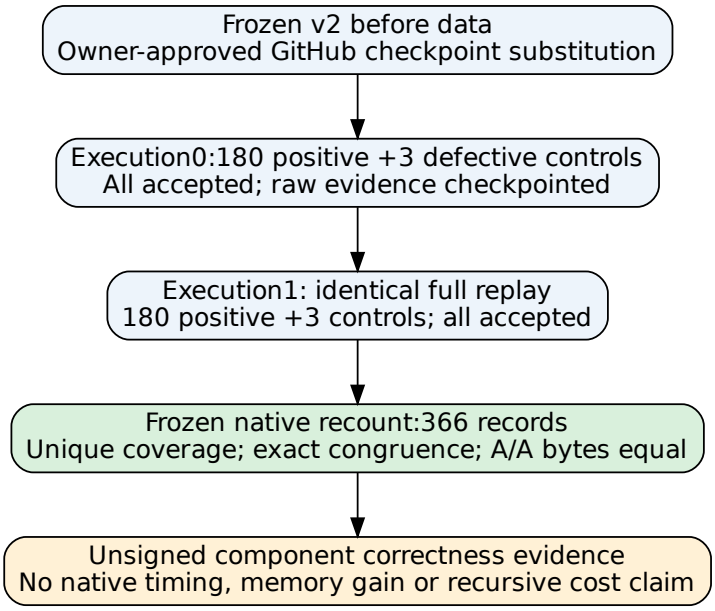
Flow:2 planned,2 attempted,2 completed;0 interrupted/resumed/excluded. Complete deterministic census: no statistical interval or population inference.



Workload. Five triples(a, u, M), three immutable sparse weight fixtures, two injective placements, output precisions1/4/9 and canonical/offset lifts give180 positive cases. The $a = 4$ domain is held out relative to $a = 3$ development. All actual active source nodes and original child precisions are used. Exact source and binary hashes are frozen; C++20/GMP6.3.0, g++14.2.0.

Controls. Zero quotient misses delta target1; corrupting node1's quotient constant by one yields nonintegrality; a scalar affine control separates $D - N = 1$ from $1 - N = -1$ at $D = 3, N = 2$. The affine control tests that identity, not an entire faulty contraction implementation. Full second invocation is A/A; exact byte spread zero.

Methods, provenance and limits



Instrument. Source `77aed44a5`; unchanged checker and analyzer. The checker uses actual affine clearing, dense small-fixture polynomial charts, finite modular inverse/convolution, integer lifts and a rational Hermite sum. The analyzer uses numerator-minus-target-times-denominator divisibility, separately from the checker's denominator inversion.

Admission. Owner approved GitHub checkpoints instead of unavailable S3. Exact protocolv1 is retained; v2 changes only checkpoint transport before data. Freeze20:07:22UTC; lifecycle markers run0 20:07:42–20:07:49, run1 20:07:53–20:08:00. These are provenance, not timing measurements. Local Linux x86-64/AMD EPYC9V74 model; each single-thread checker pinned to CPU0,2GiB address-space cap,600s timeout,32MiB output cap; no retries. Analyzer1CPU/2GiB/60s. Host co-tenants not isolated, L0, clocks invalid.

Evidence. Raw panels and receipts in `../../runs/`; native recount `../../summary.json`; canonical two-row `../../export.json`. Schema and native artifact bindings pass. Both raw TSV SHA256 values: `46a73583d9709c64acbcf02fbadb82734ed48b16cb5c33d85341ca74db512c09`. Protocol hash: `7d5b77252d2930a0e8da185f08e0454809af934270dfa94c1a1a44bd50d57635`.

Scope. Finite native component correctness only. No timing, measured memory-saving, full recursive runtime or curve solve result. Same-producer recount/replay is not independent signed verification. Next: native refinement of the distinct modular accumulator under a prospective amendment. Sources are actual formal recovery and pinned manuscript D.9/D.10; KB unavailable.